

Lab 1 Revision Questions

Medical Technology & Laboratory Safety

I. MULTIPLE CHOICE QUESTIONS (MCQS)

1. Which device uses pressurized steam at 121°C to 134°C for sterilization?

- a) Centrifuge b) Autoclave c) Incubator d) Microscope

Answer: b) Autoclave

2. The primary purpose of a Centrifuge is to separate liquid components based on:

- a) Color b) Volume c) Density d) Temperature

Answer: c) Density

3. Which type of pipette is used for transferring very precise microliter volumes of liquid?

- a) Pasteur Pipette b) Graduated Pipette c) Micropipette d) Volumetric Pipette

Answer: c) Micropipette

4. A Spectrophotometer is used to measure the amount of _____ absorbed by a chemical substance.

a) Heat b) Light c) Pressure d) Electricity

Answer: b) Light

5. Which equipment provides a controlled environment (Temp, Humidity, CO2) for cell growth?

a) Fume Hood b) Biosafety Cabinet c) Incubator d) Shaker

Answer: c) Incubator

6. To diagnose bleeding disorders and monitor anticoagulant therapy, we use a:

a) Hematology Analyzer b) Coagulation Analyzer c) Chemistry Analyzer d) pH Meter

Answer: b) Coagulation Analyzer

7. Class III Biosafety Cabinets are designed for handling pathogens of which risk level?

a) BSL-1 b) BSL-2 c) BSL-3 d) BSL-4

Answer: d) BSL-4

8. Personal Protective Equipment (PPE) includes all the following EXCEPT:

a) Gloves b) Lab coats c) Fume hoods d) Safety goggles

Answer: c) Fume hoods (It is an Engineering Control)

9. Which device is specifically used to count and identify blood cells (CBC)?

- a) Biochemistry Analyzer b) Hematology Analyzer c) Blood Gas Analyzer d) Immunoassay Analyzer

Answer: b) Hematology Analyzer

10. The main function of a Fume Hood is to:

- a) Sterilize samples b) Protect the sample from contamination c) Extract dangerous fumes and gases d) Incubate bacteria

Answer: c) Extract dangerous fumes and gases

II. TRUE OR FALSE QUESTIONS

1. Analytical Balances are used to measure the mass of an object with a high degree of precision.

Answer: True

2. Class I Biosafety Cabinets (BSC) provide protection for both the user and the sample.

Answer: False (They protect the user/environment but NOT the sample)

3. An Autoclave typically runs for 15-20 minutes to achieve complete sterilization.

Answer: True

4. Safety goggles should only be worn when working with strong acids.

Answer: False (They should be worn whenever there is a risk of splashes or debris)

5. Immunoassay Analyzers use antibodies as reagents to detect specific substances.

Answer: True

6. Lab coats are considered "Emergency Safety Equipment".

Answer: False (They are part of Personal Protective Equipment - PPE)

7. Automated microbiology analyzers provide results faster than traditional manual methods.

Answer: True

8. A pH meter is used to measure the acidity or alkalinity of a solution.

Answer: True

9. Blood Gas Analyzers are used to measure electrolytes and metabolites like glucose.

Answer: False (They measure pH, pO₂, and pCO₂; Biochemistry analyzers measure glucose)

10. Flammable materials should be stored in regular open shelves for easy access.

Answer: False (They must be stored in secure, labeled chemical storage cabinets)

III. FILL IN THE BLANKS

1. The _____ is a microscope that uses a beam of electrons to create an image of the specimen.

Answer: Electron Microscope

2. PPE stands for _____ Equipment.

Answer: Personal Protective

3. The device used to rinse eyes immediately after chemical exposure is the _____ station.

Answer: Eye wash

4. _____ cabinets are specialized units to prevent the escape of biological agents.

Answer: Biosafety (or Biological Safety)

5. A _____ is used to measure the mass of very small samples with 4-decimal place accuracy.

Answer: Analytical Balance

6. Centrifugation separates components by using _____ force.

Answer: Centrifugal

7. Sharp objects like needles should be disposed of in a designated _____.

Answer: Sharps Container

8. A _____ incubator is used to provide continuous motion for cell culture growth.

Answer: Shaking

9. The _____ analyzer is used to identify blood types and cross-match for transfusions.

Answer: Immunohematology (or Blood Bank analyzer)

10. _____ are ventilated workstations that extract dangerous gases away from the user.

Answer: Fume Hoods
